

THE HYPOTHALAMUS

The hypothalamus, which literally means “below the thalamus”, is about the size of a sugar cube and contains numerous tiny clusters of neurons called nuclei. It forms an important part of the relationship between the brain and the body, and is usually regarded as the vital integrating centre of the limbic system. A stalk below links it to the pituitary gland (see p.134), which helps to regulate the activity of the endocrine system, including the thyroid and adrenal glands. The hypothalamus

also has complex associations with the rest of the limbic system around it, and with the autonomic parts of the general nervous system. Functions of the hypothalamus include monitoring and regulating vital internal conditions such as nutrient levels, body temperature, water–salt balance, blood flow, the sleep–wake cycle, and the levels of hormones such as sex hormones. The hypothalamus also initiates feelings, actions, and emotions such as hunger, thirst, rage, and terror.

HYPOTHALAMIC NUCLEI

The roles played by all the nuclei in the hypothalamus are not fully known. However, some roles have been identified. For example, the ventromedial nucleus is responsible for feelings of fullness after eating. Damage to this area causes overeating.

